

Project-MTH210
Statistical Computing

Due date is 23.04,2026, 7 p.m. No extension and no consultation with your friends. If I find similar copies all will get zero mark. It has to be submitted in typed hard copy maximum page limit is 4 (two pages printed both sides). Just staple it and put it in my mailbox. Do not use any folder. All the codes you submit it through e-mail to: kundu.debasis@gmail.com. The codes should be verifiable. If the code does not give the results you had submitted you will get zero. For the data set, send an e-mail to me (kundu@iitk.ac.in) and you will get the data set within 48 hours.

1. Consider the data set you receive in your e-mail. The data set has been generated from the following PDF with $\alpha > 0$ and $\lambda > 0$.

$$f(x; \alpha, \lambda) = \alpha \lambda e^{-\lambda x} (1 - e^{-\lambda x})^{\alpha-1}; \quad \text{for } x > 0,$$

and 0, otherwise.

Find the MLEs of α and λ based on Newton Raphson (NR) algorithm and also calculate 95% confidence intervals of both α and λ based on parametric and non-parametric bootstrap methods.

While you are writing the report, please mention the following in details.

1. Mention the file number you have received.
2. Mention the initial guess values you have used for the NR algorithm.
3. Mention what type of stopping criterion you have used.
4. Mention in how many iterations the algorithm converges.
5. Mention how many replications you have used for boot strapping.